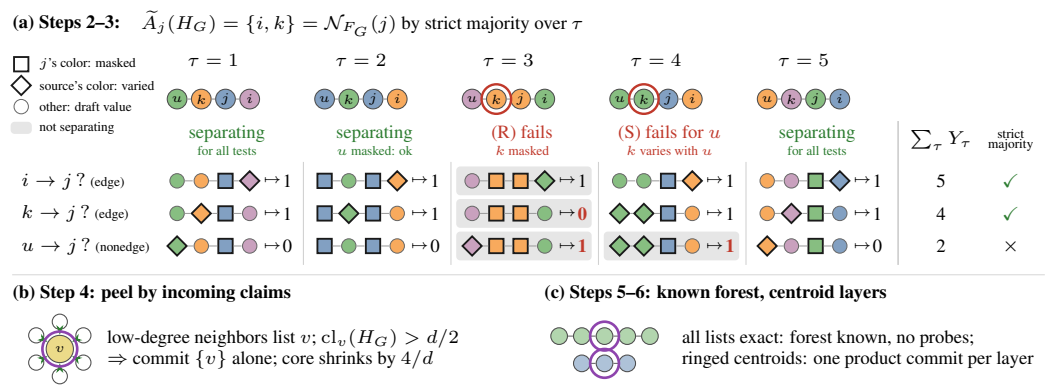

(a) Steps 2–3: $\widetilde{A}_j(H_G) = \{i,k\} = \mathcal{N}_{F_G}(j)$ by strict majority over τ
j's color: masked
source's color: varied
other: draft value
not separating
$\tau = 1$
$\tau = 2$
$\tau = 3$
$\tau = 4$
$\tau = 5$
separating
for all tests
separating
u masked: ok
(R) fails
k masked
(S) fails for u
k varies with u
separating
for all tests
$\sum_\tau Y_\tau$
strict majority
$i \to j$? (edge)
$k \to j$? (edge)
$u \to j$? (nonedge)
5
4
2
(b) Step 4: peel by incoming claims
low-degree neighbors list v; $\mathrm{cl}_v(H_G) > d/2$
$\Rightarrow$ commit $\{v\}$ alone; core shrinks by $4/d$
(c) Steps 5–6: known forest, centroid layers
all lists exact: forest known, no probes;
ringed centroids: one product commit per layer